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RYERSON UNIVERSITY

Ryerson University

Department of Mechanical and Industrial Engineering

MTL200

TERM TEST 2 (Closed Book)

Prof: C. (Ravi) Ravindran

March 25, 2011

Duration: 50 minutes

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Name: (First Name) [redacted] **(Last Name)** [redacted]

Section: 14

Student Number: [redacted]

Question	1	2	3	4	5	Total
Max. Marks	10	10	10	10	10	50
Marks Obtained	6	6	5	10	10	37

Q1: Multiple Choice (Only one correct answer please)

1. A metastable state or microstructure is
   a) Stable state
   b) Nonequilibrium state
   c) Only temporary
   d) Gaseous

2. A phase diagram with complete liquid and solid solubility of the two components
   a) Eutectic
   b) Peritectic
   c) Isomorphous
   d) Monotectic

3. In practical solidification situations, compositional readjustments and maintenance of equilibrium are not possible because of
   a) Rapid cooling rates
   b) Slow cooling rates
   c) High temperatures
   d) Gravitational effects

4. In preheating of a casting cored structure, the first entity to melt will be
   a) Centre of the grain
   b) Grain boundary
   c) Solute particles
   d) Solvent particles

5. Compounds with distinct chemical formulas are
   a) Intermetallic compounds
   b) Terminal solid solutions
   c) Intermediate solid solutions

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d) Insoluble compounds

6. In steriosomers, R groups on alternate sides of the chain in
   a) Atactic configuration
   b) Syndiotactic configuration
   c) Isotactic configuration
   d) Random configuration

7. Side branches in polymers
   a) Lower polymer density
   b) Increase polymer density
   c) Do not affect polymer density
   d) Homogenize polymer density

8. Side branches
   a) Promote crystallization
   b) Promote chain alignment
   c) Prevent crystallation
   d) Align into crystalline structure

9. Higher strength and resistance to dissolution and heat are associated with
   a) Amorphous polymers
   b) Partially amorphous polymers
   c) Thermoplastic polymers
   d) Crystalline polymers

10. Any chain disorder or misalignment in a polymer will result in
   a) Amorphous region
   b) Partially amorphous polymers
   c) Thermoelastic polymers
   d) Crystalline region

Q2: A 1.5-kg specimen of an 90 wt% Pb–10 wt% Sn alloy is heated to 250°C (480°F); at this temperature it is entirely an α-phase solid solution. The alloy is to be melted to the extent that 50% of the specimen is liquid, the remainder being the α phase. This may be accomplished by heating the alloy or changing its composition while holding the temperature constant. (Check Pb-Sn Phase Diagram Attached)

(a) To what temperature must the specimen be heated?

(b) How much tin must be added to the 1.5-kg specimen at 250°C to achieve this state?

$m = 1.5kg$ $\quad L = \alpha = 50\%$

a. $T = 290°C$ (4) $\quad T = 295°C$

b. $m_{Sn} = ?$

$C_\alpha = 16$ wt% Sn (14 wt%) $\quad C_L = 45$ wt% Sn (34 wt%)

$m_0 = (0.10)(1.5kg)$
$= 0.15kg$ ✓
$m_T = 1.5kg$ ✓

$$\frac{m_0 + m_{Sn}}{m_T + m_{Sn}} \times 100 = C_0$$

$$W_\alpha = \frac{C_L - C_0}{C_L - C_\alpha} \qquad W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha}$$

$$0.5 = \frac{45 - C_0}{45 - 16} \qquad 0.5 = \frac{C_0 - 16}{45 - 16}$$

$C_0 = 30.5$ $\quad = \quad$ $C_0 = 30.5$ (2)

$$30.5 = 100 \times \frac{0.15kg + m_{Sn}}{1.5kg + m_{Sn}}$$

$0.4575kg + 0.305 m_{Sn} = 0.15kg + m_{Sn}$

$$m_{Sn} = \frac{0.3075kg}{0.695}$$

$= 0.44245 kg = 0.44 kg$ of tin 0.276 kg

Q3: For a copper–silver alloy of composition 25 wt% Ag–75 wt% Cu and at 775°C (1425°F), do the following (Use Ag-Cu phase diagram attached):

(a) Determine the mass fractions α of β and phases.

(b) Determine the mass fractions of primary α and eutectic microconstituents.

(c) Determine the mass fraction of eutectic α.

a. $W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha}$ ✓

$= \frac{91.2 - 25}{91.2 - 8.0}$

$= 0.7957$ ✓

$= 0.80$

$W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha}$ ✓

$= \frac{25 - 8.0}{91.2 - 8.0}$

$= 0.2043$ ✓

$= 0.20$

4

b. $W_{\alpha'} = \frac{C_\beta - C_E}{C_\beta - C_\alpha}$ X $= \frac{C_E - C_0}{C_E - C_\alpha}$

$= \frac{91.2 - 71.9}{91.2 - 8.0}$ $= 0.234$

$= 0.23197$ (−2)

$= 0.23$ X

$W_E = \frac{C_E - C_\alpha}{C_\beta - C_\alpha}$ X $= \frac{C_0 - C_\alpha}{C_E - C_\alpha}$

$= 0.266$

$= \frac{71.9 - 8.0}{91.2 - 8.0}$

(−2)

$= 0.76803$

$= 0.77$ X

c. $W_{eutectic\,\alpha} = W_\alpha - W_{\alpha'}$ ✓

$= 0.7957 - \cancel{0.23197}$ 0.734

$= \cancel{0.56373}$ $= 0.062$

$= 0.56$ X (−1)

5

1

/5

Q4: Calculate the number-average molecular weight of a random nitrile rubber [poly(acrylonitrile-butadiene) copolymer] in which the fraction of butadiene repeat units is 0.30; assume that this concentration corresponds to a degree of polymerization of 2000.

$f_{BU} = 0.30$

$f_{AC} = 1 - f_{BU} = 0.70$

$m_{BU} = 4C + 6H$

$= 4(12.01\,g/mol) + 6(1.008\,g/mol)$

$= 54.088\ g/mol$

$m_{AC} = 3C + 3H + N$

$= 3(12.01\,g/mol) + 3(1.008\,g/mol) + 14.01\,g/mol$

$= 53.064\ g/mol$

$\bar{m} = \sum m_i f_i$

$= m_{BU} f_{BU} + m_{AC} f_{AC}$

$= (54.088\,g/mol)(0.30) + (53.064\,g/mol)(0.70)$

$= 53.3712\ g/mol$

$DP = \frac{\bar{M}_n}{\bar{m}}$

$\bar{M}_n = DP \cdot \bar{m}$

$= 2000 \cdot (53.3712\,g/mol)$

$= 106\ 742.4\ g/mol$

$= 1.1 \times 10^5\ g/mol$

10/10

Q5: The density and associated percent crystallinity for two polytetrafluoroethylene materials are as follows:

ρ(g/cm^3)	Crystallinity (%)
2.144	51.3
2.215	74.2

(a) Compute the densities of totally crystalline and totally amorphous polytetrafluoroethylene.

(b) Determine the percent crystallinity of a specimen having a density of 2.26 g/cm^3.

a. Material 1:

$$\frac{C_1}{100} = \frac{\rho_c(\rho_{s_1} - \rho_a)}{\rho_{s_1}(\rho_c - \rho_a)}$$

$$C_1\rho_{s_1}\rho_c - C_1\rho_{s_1}\rho_a = \rho_c\rho_{s_1} - \rho_c\rho_a$$

$$\rho_{s_1}\rho_c(C_1 - 1) = \rho_a(C_1\rho_{s_1} - \rho_c)$$

$$\rho_a = \frac{\rho_{s_1}\rho_c(C_1 - 1)}{(C_1\rho_{s_1} - \rho_c)} \quad (1)$$

Material 2:

$$\frac{C_2}{100} = \frac{\rho_c(\rho_{s_2} - \rho_a)}{\rho_{s_2}(\rho_c - \rho_a)}$$

$$\rho_{s_2}\rho_c(C_2 - 1) = \rho_a(C_2\rho_{s_2} - \rho_c) \quad (2)$$

Sub (1) in (2)

$$\rho_{s_2}\rho_c(C_2 - 1) = \frac{\rho_{s_1}\rho_c(C_1 - 1)}{(C_1\rho_{s_1} - \rho_c)}(C_2\rho_{s_2} - \rho_c)$$

$$2.215\rho_c(0.742 - 1) = \frac{2.144 \cdot \rho_c(0.513 - 1)}{0.513 \cdot 2.144 - \rho_c}(0.742 \cdot 2.215 - \rho_c)$$

$$1.64353\rho_c - 2.215\rho_c = \frac{(1.0999\rho_c - 2.144\rho_c)}{1.0999 - \rho_c}(1.6435 - \rho_c)$$

$$-0.6286\rho_c + 0.57147\rho_c^2 = -1.7160\rho_c + 1.0441\rho_c^2$$

$$1.0874 = 0.47263\rho_c$$

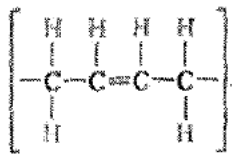
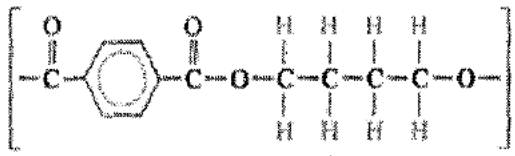
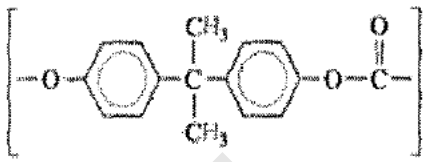
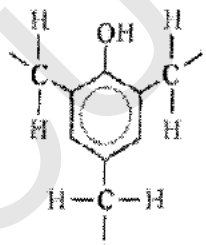
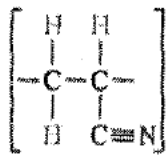
$$\rho_c = 2.3007 \text{ g/cm}^3$$

$$= 2.30 \text{ g/cm}^3$$

$$\rho_a = \frac{(2.144)(2.3007)(0.513 - 1)}{(0.513 \cdot 2.144 - 2.3007)}$$

$$= 2.0005 \text{ g/cm}^3$$

$$= 2.00 \text{ g/cm}^3$$

Chemical Name	*Repeat Unit Structure*
Polybutadiene	
Poly(butylene terephthalate) (PBT)	
Polycarbonate (PC)	
Phenol-formaldehyde (phenolic)	
Polyacrylonitrile (PAN)	

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Atomic Weights (g/mol)

C	12.01
N	14.01
H	1.008

Formula Sheet

$$\%\ Crystallinity = \frac{\rho_c(\rho_s - \rho_a)}{\rho_s(\rho_c - \rho_a)} \times 100\%$$

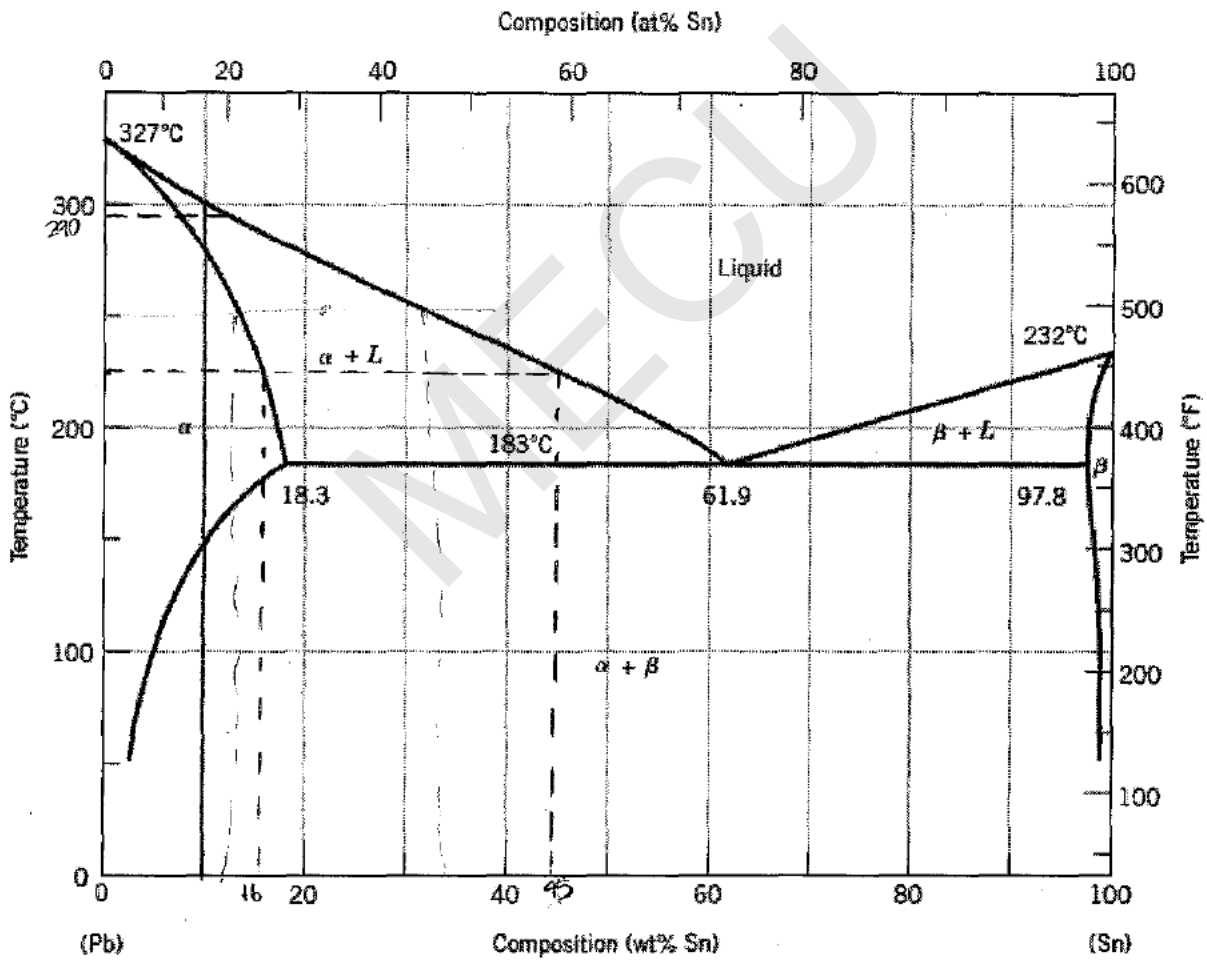

Figure 9.8 The lead–tin phase diagram. [Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 3, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

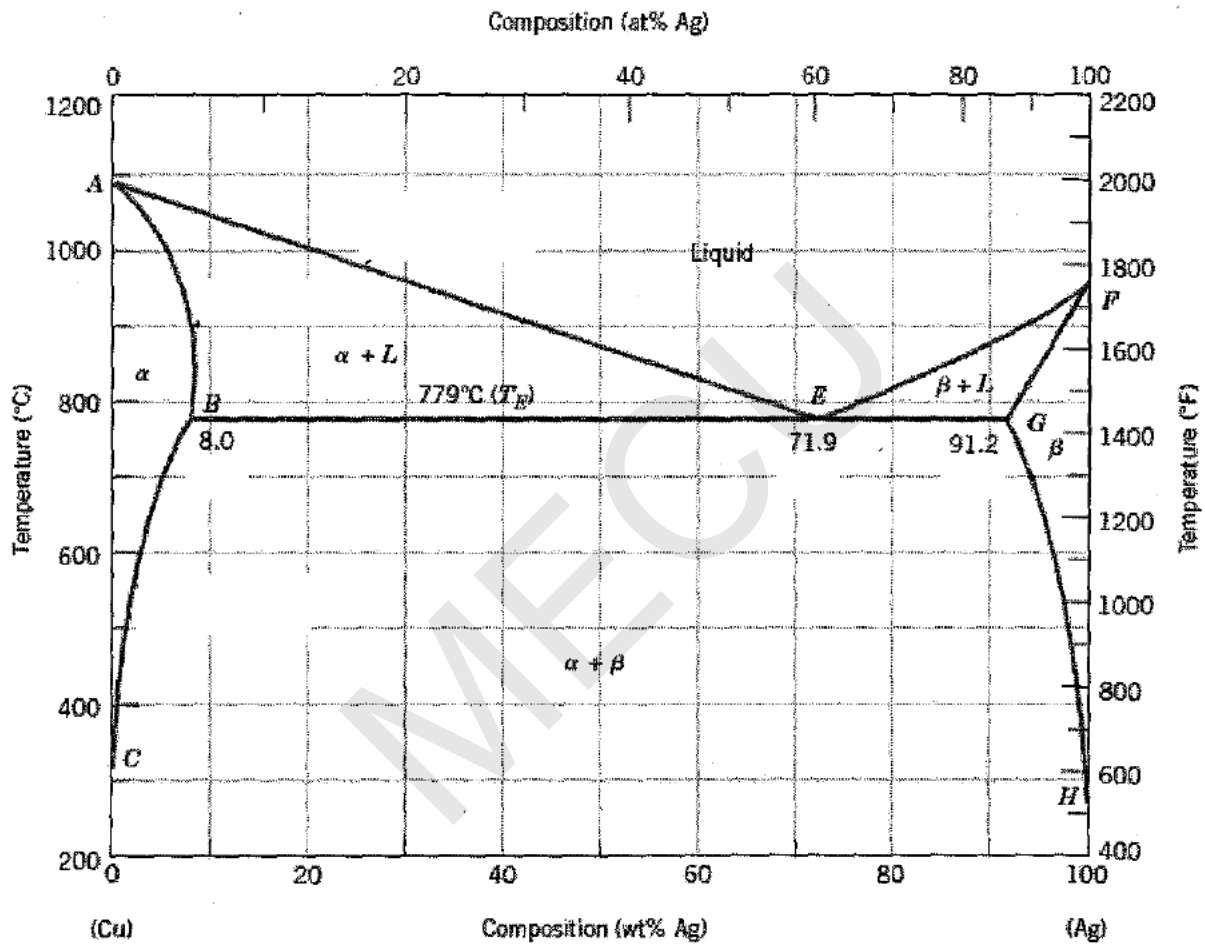

Figure 9.7 The copper–silver phase diagram. [Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 1, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]